Jonathan Huecker
Vice President
Catawba Nuclear Station

Duke Energy
CN01VP | 4800 Concord Road
York, SC 29745

o: 803.701.3349

RA-25-0222

10 CFR 50.73

September 15, 2025

U.S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, D.C. 20555

Subject: Duke Energy Carolinas, LLC
Catawba Nuclear Station, Units 1 and 2
Docket No. 50-413, 50-414
Licensee Event Report (LER) 2025-003-00

Pursuant to 10 CFR 50.73(a)(1) and (d), attached is LER 414/2025-003-00, entitled "2B Emergency Diesel Generator Field Failed to Flash". This report is being submitted in accordance with 10 CFR 50.73(a)(2)(i)(B) and 10 CFR 50.73(a)(2)(v)(D).

There are no regulatory commitments contained in this letter or its attachment.

The impact of this event in relation to plant risk was low and there was no impact to the health and safety of the public.

If questions arise regarding this LER, please contact Ari Tuckman of Regulatory Affairs at (803) 701-3771.

Sincerely,

Jonathan Huecker
Vice President, Catawba Nuclear Station

Attachment

xc (with attachment):

Julio Lara
Acting Regional Administrator
U.S. Nuclear Regulatory Commission - Region II

Jack Minzer-Bryant
NRC Project Manager (CNS)
U.S. Nuclear Regulatory Commission

David Rivard
Senior Resident Inspector, Catawba Nuclear Station
U.S. Nuclear Regulatory Commission

NRC FORM 366 (04-02-2024) U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104 EXPIRES: 04/30/2027
LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)	Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name		2. Docket Number	3. Page
Catawba Nuclear Station, Unit 2	[X] 050 [ ] 052	00414	1 of 5

4. Title
2B Emergency Diesel Generator Field Failed to Flash

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name		Docket Number
07	15	2025	2025	- 003 -	00	09	15	25	Catawba Unit 1	[X] 050	00413
									Facility Name	[ ] 052	Docket Number

9. Operating Mode	10. Power Level
1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	[ ] 20.2203(a)(2)(vi)	**10 CFR Part 50**	[ ] 50.73(a)(2)(ii)(A)	[ ] 50.73(a)(2)(viii)(A)	[ ] 73.1200(a)
[ ] 20.2201(b)	[ ] 20.2203(a)(3)(i)	[ ] 50.36(c)(1)(i)(A)	[ ] 50.73(a)(2)(ii)(B)	[ ] 50.73(a)(2)(viii)(B)	[ ] 73.1200(b)
[ ] 20.2201(d)	[ ] 20.2203(a)(3)(ii)	[ ] 50.36(c)(1)(ii)(A)	[ ] 50.73(a)(2)(iii)	[ ] 50.73(a)(2)(ix)(A)	[ ] 73.1200(c)
[ ] 20.2203(a)(1)	[ ] 20.2203(a)(4)	[ ] 50.36(c)(2)	[ ] 50.73(a)(2)(iv)(A)	[ ] 50.73(a)(2)(x)	[ ] 73.1200(d)
[ ] 20.2203(a)(2)(i)	**10 CFR Part 21**	[ ] 50.46(a)(3)(ii)	[ ] 50.73(a)(2)(v)(A)	**10 CFR Part 73**	[ ] 73.1200(e)
[ ] 20.2203(a)(2)(ii)	[ ] 21.2(c)	[ ] 50.69(g)	[ ] 50.73(a)(2)(v)(B)	[ ] 73.77(a)(1)	[ ] 73.1200(f)
[ ] 20.2203(a)(2)(iii)		[ ] 50.73(a)(2)(i)(A)	[ ] 50.73(a)(2)(v)(C)	[ ] 73.77(a)(2)(i)	[ ] 73.1200(g)
[ ] 20.2203(a)(2)(iv)		[X] 50.73(a)(2)(i)(B)	[X] 50.73(a)(2)(v)(D)	[ ] 73.77(a)(2)(ii)	[ ] 73.1200(h)
[ ] 20.2203(a)(2)(v)		[ ] 50.73(a)(2)(i)(C)	[ ] 50.73(a)(2)(vii)		

[ ] **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Ari Tuckman, Lead Engineer, CNS Regulatory Affairs	(803) 701-3771

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	EK	CON	S375	Y					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
[X] No	[ ] Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On 7/15/2025, Catawba Unit 2 was in Mode 1 operating at approximately 100 percent power. At 8:34 EDT the 2B Emergency Diesel Generator (DG)[EK] was manually started for scheduled testing. Upon start, voltage and frequency parameters indicated the 2B DG field did not flash. Investigation determined a sliding link in the field flash circuit was not in the closed position. This portion of the system energizes the field flash contactor to allow voltage buildup during 2B DG startup.

The previous successful operation of the 2B DG was on 7/1/2025 for an approximate 12 hour run. During the time between 7/1/2015 and 7/15/2025 there were two instances when the 2A DG was removed from service. This condition resulted in both an “operation or condition prohibited by Technical Specifications” reportable under 10 CFR 50.73(a)(2)(i)(B) and a “condition that could have prevented fulfillment of a safety function” reportable under 10 CFR 50.73(a)(2)(v)(D).

There was no impact on the health and safety of the public.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
			YEAR	SEQUENTIAL NUMBER	REV NO.
Catawba Nuclear Station, Units 1 and 2	[X] 050 / [ ] 052	00413, 00414	2025	003	00

NARRATIVE

BACKGROUND

The following information is provided to assist readers in understanding the event described in this LER. Applicable Energy Industry Identification System [EIIS] and component codes are enclosed within brackets. Catawba's unique system and component identifiers are contained within parentheses.

The onsite standby power source for each 4160 volt Engineered Safety Features (ESF) bus at Catawba is a dedicated DG. For each unit, DGs A and B are dedicated to ESF buses ETA and ETB, respectively. Each DG starts automatically on a Safety Injection (SI) signal (i.e., low pressurizer pressure or high containment pressure) or on an ESF bus degraded voltage or undervoltage signal. After the DG has started, it will automatically tie to its respective bus after offsite power is tripped as a consequence of ESF bus undervoltage or degraded voltage, independent of or coincident with an SI signal. The DGs will also start and operate in the standby mode without tying to the ESF bus on an SI signal alone. Following the trip of offsite power, a sequencer strips loads from the ESF bus. When the DG is tied to the ESF bus, loads are then sequentially connected to its respective ESF bus by the automatic load sequencer. In the event of a loss of preferred power, the ESF electrical loads are automatically connected to the DGs in sufficient time to provide for safe reactor shutdown and to mitigate the consequences of a Design Basis Accident such as a Loss of Coolant Accident (LOCA).

There are also provisions to accommodate the connecting of the Emergency Supplemental Power Source (ESPS) to one train of either unit's Class 1E AC Distribution System. The ESPS consists of two 50% capacity non-safety related commercial grade DGs. Manual actions are required to align the ESPS to the station and only one of the station's four onsite Class 1E Distribution System trains can be supplied by the ESPS at any given time. The ESPS is made available to support extended Completion Times in the event of an inoperable DG as well as a defense-in-depth source of AC power to mitigate a station blackout event. The ESPS remains disconnected from the Class 1E AC Distribution System unless required for supplemental power to one of the four 4.16 kV ESF buses.

Technical Specifications:

Catawba TS 3.8.1 governs the DGs for each unit that is in Modes 1, 2, 3, and 4.

Limiting Condition for Operation (LCO) 3.8.1(b) requires two operable DGs capable of supplying the Onsite Essential Auxiliary Power Systems. With one LCO 3.8.1(b) DG inoperable, the inoperable DG must be restored to operable status within 72 hours per Required Action B.6.

LCO 3.8.1(d) requires two DGs from the opposite unit necessary to supply power to the shared systems and the Nuclear Service Water System (NSWS) pump(s). With one LCO 3.8.1 (d) DG inoperable, the inoperable DG must be restored to operable status within 72 hours per Required Action D.6.

If an inoperable DG is not restored to operable status within 72 hours, the unit must be placed in Mode 3 within 6 hours and in Mode 5 within 36 hours per LCO 3.8.1 Required Actions I.1 and I.2.

In order to extend the Completion Time for an inoperable DG from 72 hours to 14 days, it is necessary to ensure the availability of the ESPS within 1 hour of entry into TS 3.8.1 LCO and every 12 hours thereafter.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Catawba Nuclear Station, Units 1 and 2	[X] 050 / [ ] 052	00413, 00414	2025	003	00

NARRATIVE

EVENT DESCRIPTION

On 7/15/2025, Catawba Unit 2 was in Mode 1 operating at approximately 100 percent power. At 8:34 EDT the 2B Emergency Diesel Generator (DG) was started for scheduled testing. Upon start of the DG, no voltage or frequency indication was observed on the Operator Aid Computer, Local or Control Room indications. The 2B DG was declared inoperable, and a troubleshooting plan was initiated to determine the cause of the 2B DG field flash failure.

Investigation identified an open electrical sliding link connection in the field flash circuit. This portion of the system energizes the field flash contactor to allow voltage buildup during 2B DG start. The sliding link was subsequently replaced and the 2B DG was started and operated as expected.

The last documented manipulation of the sliding link was on 1/16/2025 during a preventative maintenance activity to calibrate relays within the circuit. Upon conclusion of this work, the link was documented and verified as being in the closed position. Between 1/16/2025 and 7/15/2025 the 2B DG operated successfully on seven occasions for surveillance testing with no reported issues. The most recent operation of the 2B DG prior to the field flash failure on 7/15/2025 was an approximate 12 hour loaded run on 7/1/2025 with no abnormal indications. Once the engine starts and the generator field is flashed, electrical continuity through the affected sliding link is not required for the DG to continue to operate and there are no alarms or indications that would alert operators to an open circuit condition, either in operation or after the engine is secured and placed in standby alignment.

The opposite train 2A DG was considered inoperable on two occasions between 7/1/2025 and 7/15/2025. The two instances were on 7/2/2025 for less than 6 hours and again on 7/10/2025 for less than 6 hours. During these times, both emergency diesel generators on Unit 2 were inoperable which results in both an “operation or condition prohibited by Technical Specifications” reportable under 10 CFR 50.73(a)(2)(i)(B) and a “condition that could have prevented fulfillment of a Safety Function” reportable under 10 CFR 50.73(a)(2)(v)(D) for Unit 2 and Unit 1.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Catawba Nuclear Station, Units 1 and 2	[X] 050 / [ ] 052	00413, 00414	2025	003	00

NARRATIVE

CAUSAL FACTORS

The last documented manipulation of the sliding link was on 1/16/2025 during a preventative maintenance activity to calibrate relays within the circuit. The sliding link was concurrently verified in the closed position as part of the circuit restoration. Discussions post-event with technicians indicated that nothing abnormal was noted when closing the sliding link. Additionally, the technicians performed a visual inspection of the cabinet and did not identify any sliding link issues. A review of work activities was performed, and no work was identified that would have opened the link since the last successful operation of the 2B DG on 7/1/2025. The sliding link was sent to a metallurgical lab for analysis. Results of the analysis did not identify a defect that was considered likely to influence proper tightening of the sliding link.

The causal investigation determined that the most likely cause was the sliding link was not sufficiently tightened during the 1/16/25 maintenance restoration activities. The investigation was unable to determine the exact reason the sliding link was not sufficiently tightened but considered material defect causing inaccurate feedback that the sliding link was tight, foreign material within the clamping components of the sliding link, and human error.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Catawba Nuclear Station, Units 1 and 2	[X] 050 / [ ] 052	00413, 00414	2025	003	00

NARRATIVE

CORRECTIVE ACTIONS

Immediate:

1. Replace affected sliding link.
2. Validate condition (position and tightness) of sliding links in all DG control cabinets.

Planned:

1. Using performance improvement training, re-enforce verification practice of challenging sliding link tightness by attempting to move the link from the closed position.
2. Identify critical sliding links and implement the requirement for closing of sliding links to be independently verified.

SAFETY ANALYSIS

The failure to field flash of the 2B Emergency Diesel Generator presents a reduction in the reliability and availability of the Catawba Unit 2 emergency power system. There was no plant transient, radiological release, or other challenge to normal plant operation to cause any impacts to the public health and safety.

Catawba Unit 2 has a considerable level of defense-in-depth besides having two DG trains. The ESPS provides a means to restore AC power to the 2B Essential Auxiliary Power System Bus (2ETB) if the 2B DG failed to run. Similarly, another recovery action is the ability to align offsite power from Unit 1 if available via the Shared Auxiliary Transformer SATA. The Safe-Shutdown Facility (SSF) also provides an alternative means of maintaining safe shutdown conditions with the SSF Diesel Generator powering the Standby Makeup Pump for reactor coolant pump (NCP) seal cooling and instrumentation and controls for operation of Turbine-Driven Auxiliary Feedwater Pump. The final level of safety is the ability to use FLEX equipment and strategies to provide alternative core cooling to prevent core damage.

With these factors taken together there was no impact to the health and safety of the public.

ADDITIONAL INFORMATION

There have been no previous Licensee Event Reports at Catawba Nuclear Station in the last three years with the same causal factor related to sliding links.